COLLEGE OF SCIENCE
Mathematics

Parsimonious Machine Learning Methods for Time Series Forecasting of Valley Fever Infection in Maricopa County, AZ

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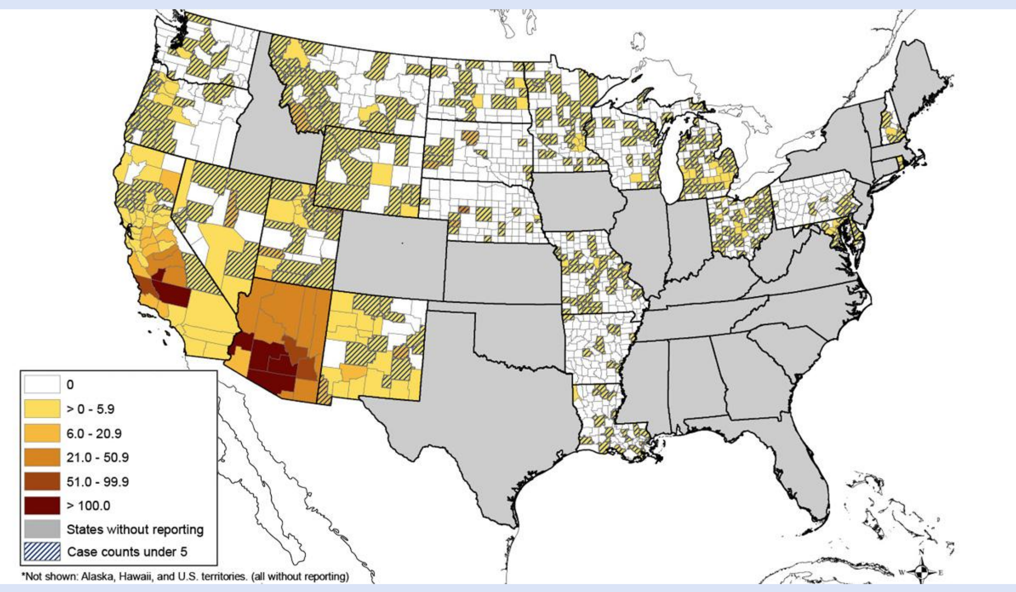
Introduction

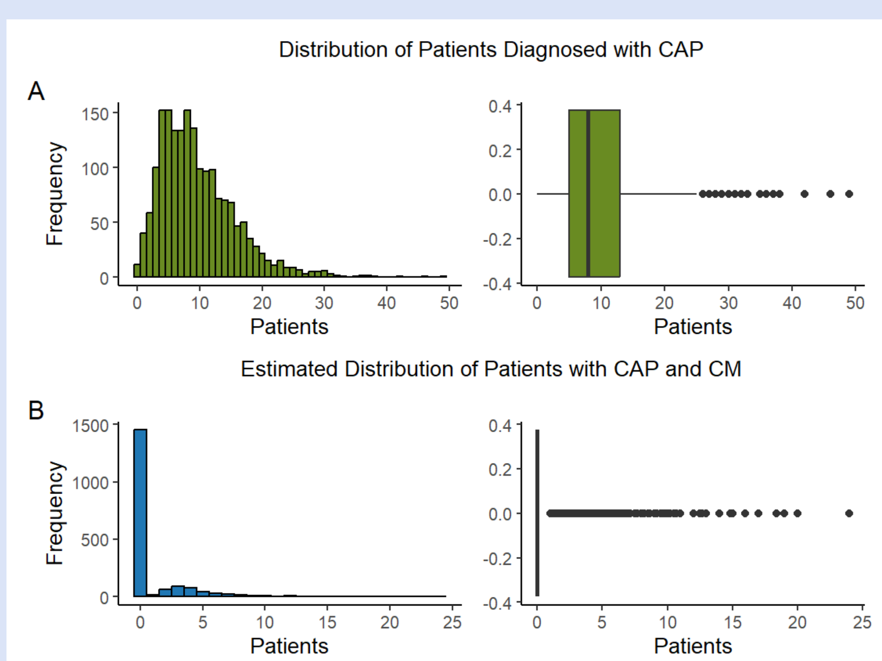
Valley fever, also known as coccidioidomycosis (CM), is a fungal infection endemic to the Southwestern U.S. caused by inhaling Coccidioides posadasii and immitis spores in soil. While CM is easy to treat with antifungal medications, it's difficult to identify in patients, and symptoms can persist for a prolonged period when left untreated. The symptoms of CM are like those of community acquired pneumonia (CAP) as both conditions are characterized by cough, fever, fatigue, and other respiratory issues.



Centers for Disease Control and Prevention (CDC). [Valley Fever incidence map]. https://www.cdc.gov/

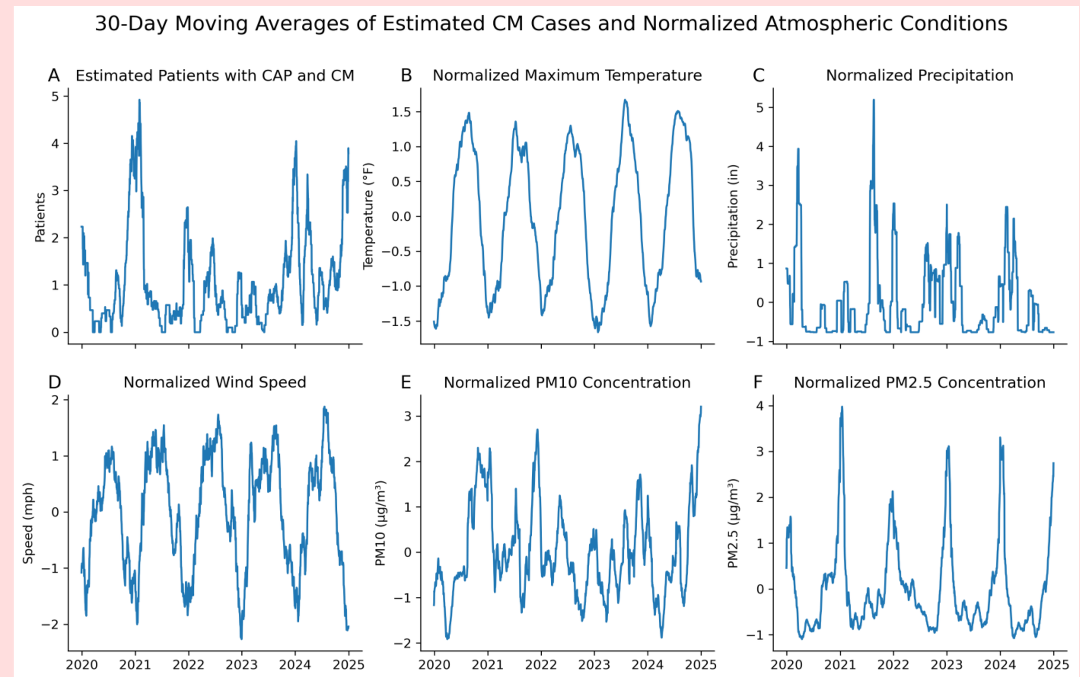
Methods

A time-series of five years of daily estimated CM cases among CAP patients at Maricopa County Banner Urgent Care Clinics (which were infrequent) and daily atmospheric and air quality data were used to forecast future CAP-associated CM infections. Extending the method of Jin et al. (2025), Long Short-Term Memory (LSTM) and Gradient Boosted Decision Tree (GBDT) models were fit exclusively to environmental factors to increase accessibility and time efficiency.

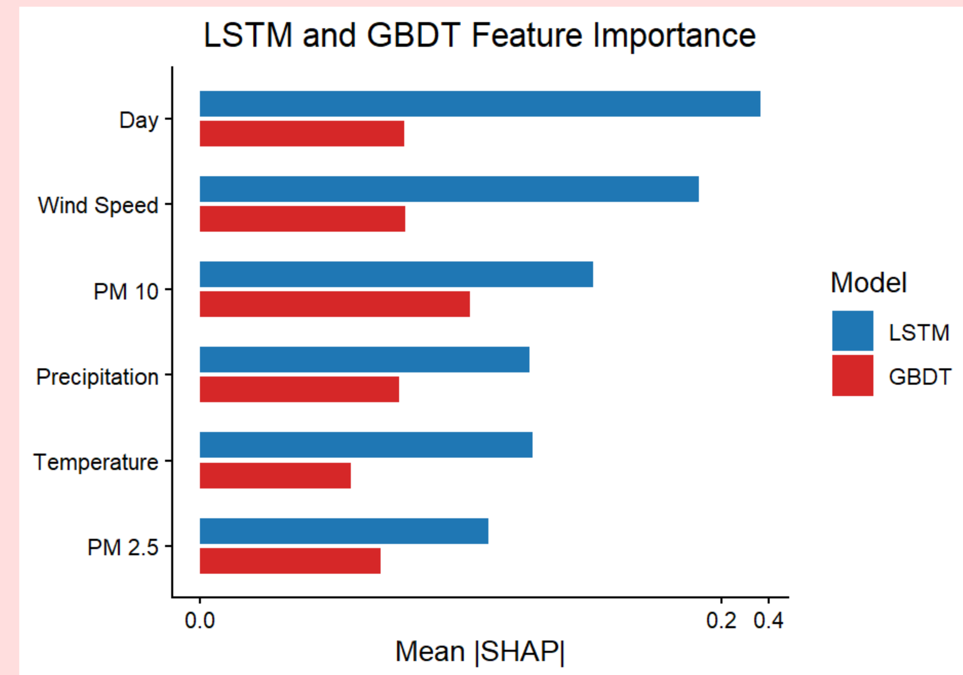


Patient distributions by diagnosis type.

Results

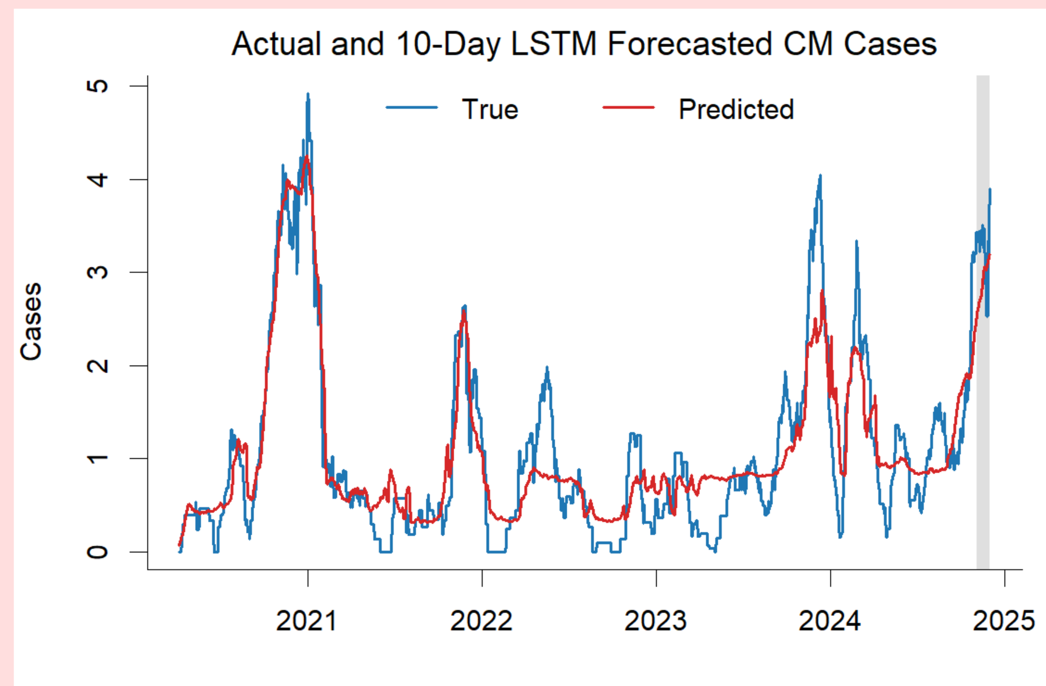


Time-series of CM cases and atmospheric conditions.

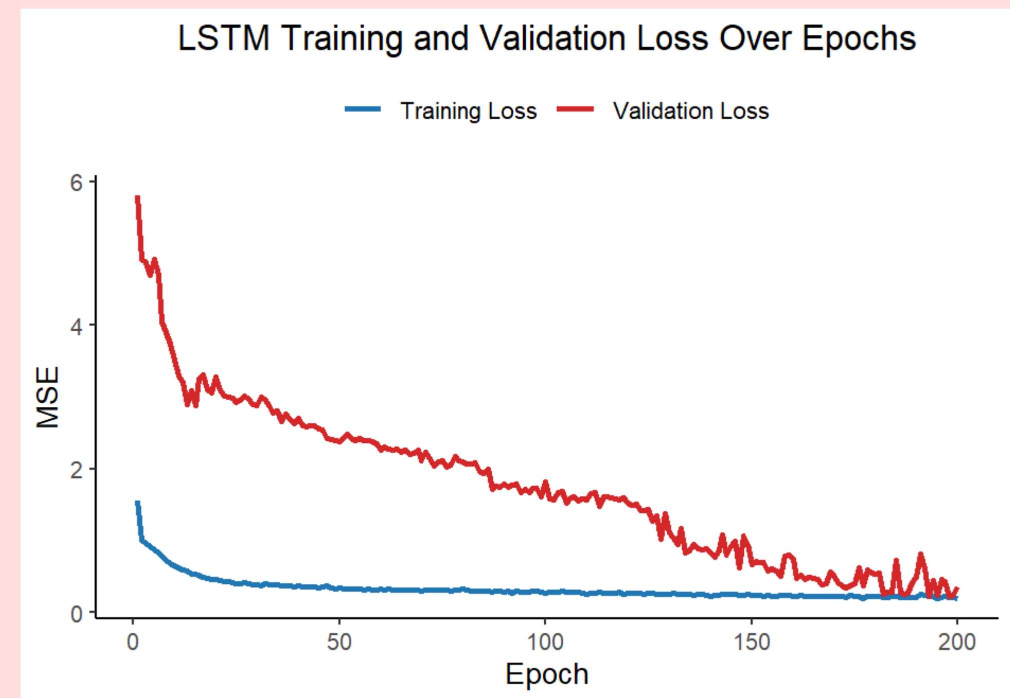


Relative contributions of features to predictions.

Long Short-Term Memory



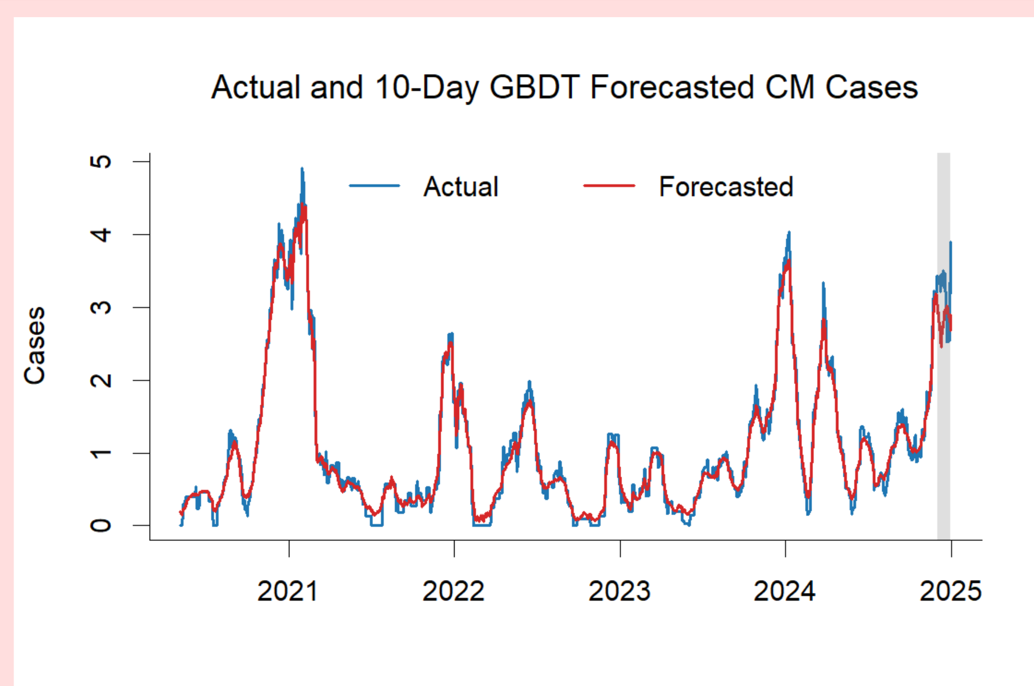
True and predicted CM cases with shaded test region.



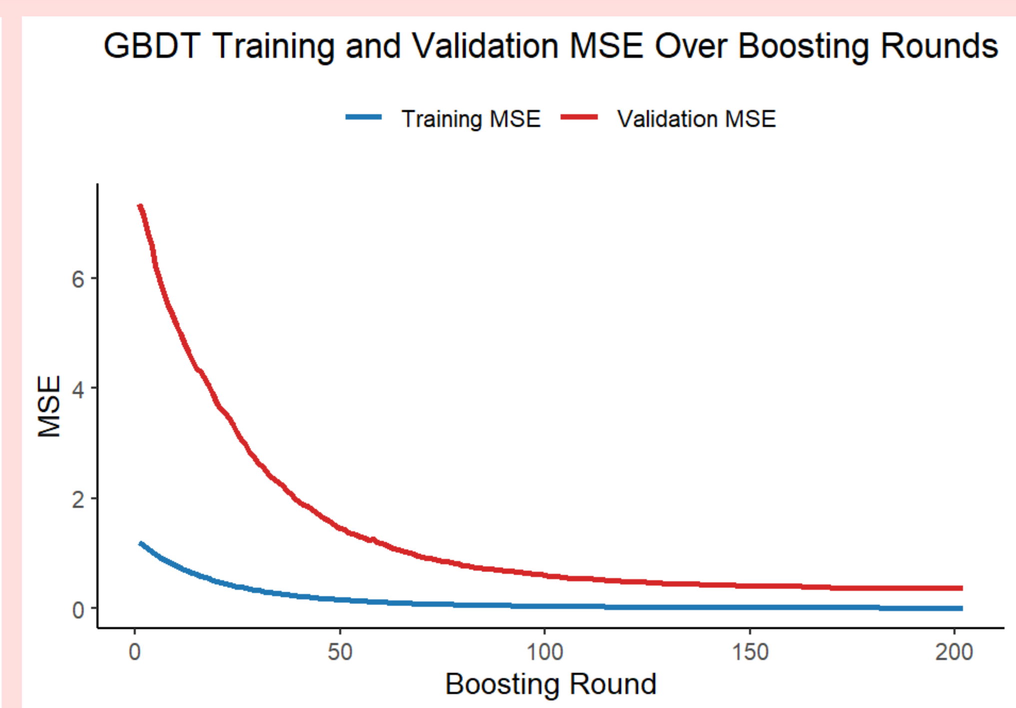
Decreasing loss over epochs indicates improving fit.

Dropout	Activation	Batch Size	Test MSE
0.1	Sigmoid	32	0.3505

Gradient Boosted Decision Tree



True and predicted CM cases with shaded test region.



Decreasing loss over rounds indicates improving fit.

η	d_{max}	w_{min}	γ	s_{row}	s_{col}	Test MSE
0.03	4	3	0	0.3	0.7	0.3699

Analysis

Although the GBDT model fits the training data more closely, the LSTM model has slightly better generalization on the thirty-day test set. The test MSEs differ only marginally, suggesting similar performances. Day dominates the LSTM model's feature importance, but feature importance is distributed more evenly for the GBDT model, with predictors contributing comparably. Computationally, the GBDT model has near-constant prediction time for fixed tree number and depth. In contrast, LSTM prediction scales with sequence length for fixed number of hidden units and input dimension.

LSTM	GBDT
$O(T(h^2 + hd))$	$O(KD)$

Worst-case prediction time complexity by model.

Conclusion

The models may be incorporated into a time-efficient real-time online tool to support healthcare providers in making informed testing decisions and motivate the public to adopt healthier practices during periods of heightened infection risk.

Yale Climate Connections. [Dust storm moving into Phoenix]. https://yaleclimateconnections.org/

Acknowledgements & Ref.

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